Chap 3. Brownian motion as a Rough Path

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Outline

1. Brownian rough path
2. Comparison of Itô and Stratonovich Brownian motion
   - Itô Brownian motion
   - Stratonovich Brownian motion
3. Brownian motion in a magnetic field
4. Scaling limits of random walks
5. Conclusion

Brownian rough path

Brownian rough path

Main object.

Random paths $X(\omega) : [0, T] \to V$ and $\mathbb{X}(\omega) : [0, T]^2 \to V \otimes V$ that are subject to the Chen's relation,

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad \forall s, u, t \in [0, T].$$

Equivalently, it is a random path

$$\mathbf{X}(\omega) = (X, \mathbb{X}) : [0, T] \to V \oplus (V \otimes V),$$

$\ast$ Check Exercise 2.7 to show that $\mathbf{X}(\omega)$ is indeed a path.

Brownian rough path

Main example.

$\mathbf{B} = (B, \mathbb{B})$,
where

B is a d-dimensional Brownian motion and is enhanced to

$\mathbb{B}_{s,t} := \int_s^t B_{s,r} \otimes dB_r \in \mathbf{R}^d \otimes \mathbf{R}^d$.

Question.

α-Hôlder regularity of X and 2α-Hôlder regularity of $\mathbb{X}$

α-Hôlder regularity of B and 2α-Hôlder regularity of $\mathbb{B}$

- Recall $B \in C^\alpha$, almost surely for any $\alpha < 1/2$.

Theorem 3.1 (Kolmogorov criterion for rough paths).

Let $q \geq 2$ and $\beta > 1/q$. For all s, t in $[0, T]$,

$$|X_{s,t}|_{L^q} \leq C|t-s|^{\beta}, \quad |\mathbb{X}_{s,t}|_{L^{q/2}} \leq C|t-s|^{2\beta}, \tag{1}$$

for some constant $C < \infty$.
Then, for all $\alpha \in [0, \beta - 1/q)$, there exists a modification of $(X, \mathbb{X})$ (also denoted by $(X, \mathbb{X})$) and random variables $K_\alpha \in L^q, \mathbb{K}_\alpha \in L^{q/2}$ such that, for all $s, t \in [0, T]$,

$$|X_{s,t}| \leq K_\alpha(w)|t-s|^{\alpha}, \quad |\mathbb{X}_{s,t}| \leq \mathbb{K}_\alpha(w)|t-s|^{2\alpha}. \tag{2}$$

Brownian rough path

Proof. WLOG let $T = 1$.
$D_n := \{\frac{k}{2^n} | k = 0, 1, ..., 2^n - 1\}$ is a partition with $|D_n| = 2^{-n}$.
We may assume that $s, t \in \cup D_n$.

$$K_n := sup_{t \in D_n} |X_{t,t+2^{-n}}|, \quad \mathbb{K}_n := sup_{t \in D_n} |\mathbb{X}_{t,t+2^{-n}}|$$

. Then, by (1),

$$E(K_n^q) \leq E(\sum_{t \in D_n} |X_{t,t+2^{-n}}|^q) \leq \sum_{t \in D_n} C^q |D_n|^{\beta q} = C^q |D_n|^{\beta q - 1},$$
$$E(\mathbb{K}_n^{q/2}) \leq E(\sum_{t \in D_n} |X_{t,t+2^{-n}}|^{q/2}) \leq \sum_{t \in D_n} C^{q/2} |D_n|^{2\beta q/2} = C^{q/2} |D_n|^{\beta q - 1}.$$

let m be the integer s.t. $|D_{m+1}| < t - s \leq |D_m|$. Then there exists a partition $s = \tau_0 < \tau_1 < ... < \tau_N = t$, where each interval $(\tau_i, \tau_{i+1}) \in D_n$ for some $n \geq m + 1$ and no three intervals have same length.

Brownian rough path

It follows that

$$|X_{s,t}| \leq \max_{0\leq i<N} |X_{s,\tau_{i+1}}| \leq \sum_{i=0}^{N-1} |X_{\tau_i,\tau_{i+1}}| \leq 2 \sum_{n\geq m+1} K_n,$$
$$|\mathbb{X}_{s,t}| \leq 2 \sum_{n\geq m+1} \mathbb{K}_n + (2 \sum_{n\geq m+1} \mathbb{K}_n)^2.$$

Then,

$$\frac{|X_{s,t}|}{|t-s|^\alpha} \leq \sum_{n\geq m+1} \frac{1}{|D_{m+1}|^\alpha} 2K_n \leq \sum_{n\geq m+1} \frac{1}{|D_n|^\alpha} 2K_n \leq 2 \sum_{n\geq 0} \frac{1}{|D_n|^\alpha} K_n =: K_\alpha,$$

where $K_\alpha \in L^q$.
This is because since $\alpha < \beta - 1/q$,

$$\|K_\alpha\|_{L^q} \leq 2 \sum_{n\geq 0} \frac{1}{|D_n|^\alpha} |E(K_n^q)|^{1/q} \leq 2 \sum_{n\geq 0} \frac{C}{|D_n|^\alpha} |D_n|^{\beta-1/q} < \infty.$$

Brownian rough path

Similarly,

$$\frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} \leq \sum_{n\geq m+1} \frac{1}{|D_{m+1}|^{2\alpha}} 2\mathbb{K}_n + (\sum_{n\geq m+1} \frac{1}{|D_{m+1}|^{\alpha}} 2K_n)^2 \leq \mathbb{K}_\alpha + K_\alpha^2,$$

where $\mathbb{K}_\alpha := 2\sum_{n\geq 0} \frac{1}{|D_n|^{2\alpha}} \mathbb{K}_n \in L^{q/2}$.
This is also because since $\alpha < \beta - 1/q$,

$$\|\mathbb{K}_\alpha\|_{L^{q/2}} \leq 2\sum_{n\geq 0} \frac{1}{|D_n|^{2\alpha}} |E(\mathbb{K}_n^{q/2})|^{2/q} \leq 2\sum_{n\geq 0} \frac{C}{|D_n|^{2\alpha}} |D_n|^{2\beta - 2/q} < \infty.$$

$\square$

Theorem 3.1 (Kolmogorov criterion for rough paths).

Let $q \geq 2$ and $\beta > 1/q$. For all s, t in $[0, T]$,

$$|X_{s,t}|_{L^q} \leq C|t-s|^{\beta}, \quad |\mathbb{X}_{s,t}|_{L^{q/2}} \leq C|t-s|^{2\beta}, \tag{3}$$

for some constant $C < \infty$.
Then, for all $\alpha \in [0, \beta - 1/q)$, there exists a modification of $(X, \mathbb{X})$ (also denoted by $(X, \mathbb{X})$) and random variables $K_\alpha \in L^q, \mathbb{K}_\alpha \in L^{q/2}$ such that, for all $s, t \in [0, T]$,

$$|X_{s,t}| \leq K_\alpha(w)|t-s|^{\alpha}, \quad |\mathbb{X}_{s,t}| \leq \mathbb{K}_\alpha(w)|t-s|^{2\alpha}. \tag{4}$$

Remark.

1. Apply $\beta = 1/2$ and all $2 < q < \infty$ to Theorem 3.1.

Then, the result holds for any $\alpha \in [0, 1/2)$.

i.e. 2α-Hôlder regularity for $\mathbb{B}$ holds almost surely for any $\alpha \in [0, 1/2)$.

2. $(B, \mathbb{B}) \in \mathscr{C}^{\alpha}$ almost surely for any $\alpha \in (1/3, 1/2)$.

In this case, $\mathbf{B} = (B, \mathbb{B})$ is called *Brownian rough path* or *enhanced Brownian motion*.

3. $(B, \mathbb{B}^{Strat}) \in \mathscr{C}_g^{\alpha}$ almost surely for any $\alpha \in (1/3, 1/2)$.

Comparison of Itô and Stratonovich Brownian motion

Itô Brownian motion

Let B be a d-dimensional Brownian motion that is enhanced to

$\mathbb{B}^{Itô}_{s,t} := \int_s^t B_{s,r} \otimes dB_r \in \mathbf{R}^d \otimes \mathbf{R}^d$,

where the stochastic integration is understood in the sense of Itô.

- We already noted that for any $\alpha \in (1/3, 1/2)$ and $T > 0$, almost surely

$$\mathbf{B} = (B, \mathbb{B}^{Itô}) \in \mathscr{C}^\alpha([0, T], \mathbf{R}^d).$$

- However, it is not a geometric rough path:

$$\mathbf{B} = (B, \mathbb{B}^{Itô}) \notin \mathscr{C}^\alpha_g([0, T], \mathbf{R}^d).$$

Itô Brownian motion

- However, it is not a geometric rough path:

$$\mathbf{B} = (B, \mathbb{B}^{It\hat{o}}) \notin \mathscr{C}_g^\alpha([0,T], \mathbf{R}^d).$$

This is because by Itô formula,

$$\begin{aligned} & d(B^i B^j) = B^i dB^j + B^j dB^i + \langle B^i, B^j \rangle dt, \quad i, j = 1, \dots, d \\ \Rightarrow \quad & Sym(\mathbb{B}^{It\hat{o}}_{s,t}) = \frac{1}{2} B_{s,t} \otimes B_{s,t} - \frac{1}{2} I(t-s) \\ & \neq \frac{1}{2} B_{s,t} \otimes B_{s,t} \end{aligned}$$

Stratonovich Brownian motion

Definition. Given continuous semimartingales M, N, the *Stratonovich integral* is

$$\int_0^t M \circ dN := \int_0^t M \; dN + \frac{1}{2}[M, N]_t$$

.

- first order product rule holds: $\quad d(MN) = M \circ dN + N \circ dM.$

Let B be a d-dimensional Brownian motion that is enhanced to

$\mathbb{B}^{Strat}_{s,t} := \int_s^t B_{s,r} \otimes dB_r \in \mathbf{R}^d \otimes \mathbf{R}^d$,

with the *Stratonovich integration*

Stratonovich Brownian motion

- For identity matrix I,

$$\mathbb{B}^{Strat}_{s,t} = \mathbb{B}^{Itô}_{s,t} + \frac{1}{2} I(t-s). \tag{5}$$

Proposition 3.5.

For any $\alpha \in (1/3, 1/2)$ and $T > 0$, almost surely

$$\mathbf{B} = (B, \mathbb{B}^{Strat}) \in \mathscr{C}^{\alpha}_{g}([0, T], \mathbf{R}^d).$$

Proof. Using the relation (5), the regularity of $\mathbf{B}$ is derived from the already established Itô case. Also, since

$$Sym(\mathbb{B}^{Itô}_{s,t}) = \frac{1}{2} B_{s,t} \otimes B_{s,t} - \frac{1}{2} I(t-s),$$

(5) yields

$$Sym(\mathbb{B}^{Strat}_{s,t}) = \frac{1}{2} B_{s,t} \otimes B_{s,t}.$$

Thus, $\mathbf{B}$ is geometric. $\square$

Stratonovich Brownian motion

Proposition 2.5.

Let $\beta \in (\frac{1}{3}, \frac{1}{2}]$. For every $(X, \mathbb{X}) \in \mathscr{C}_g^\beta([0,T], \mathbf{R}^d)$, there exists a sequence of smooth paths $X^n : [0,T] \to \mathbf{R}^d$ such that

$$(X^n, \mathbb{X}^n) := (X^n, \int_0^\cdot X^n_{0,t} \otimes dX^n_t) \to (X, \mathbb{X}) \quad \text{uniformly on } [0,T]$$

with uniform rough path bounds $sup_n \|X^n\|_\beta + \|\mathbb{X}^n\|_2\beta < \infty$. By interpolation, convergence holds in α-Hôlder rough path metric for any $\alpha \in (\frac{1}{3}, \beta)$, namely $\lim_{n\to\infty} \varrho_\alpha((X^n, \mathbb{X}^n), (X, \mathbb{X})) = 0$.

Recall: Since $(B, \mathbb{B})$ is geometric rough path, it has an approximation by smooth paths in the rough path topology, *deterministically*.

Stratonovich Brownian motion

In contrast, $(B, \mathbb{B})$ has an approximation by piecewise linear paths in rough path sense, *probabilistically*:

Proposition 3.6.

Consider dyadic piecewise-linear approximations (B^n) to B on [0,T].
That is, $B_t^{(n)} = B_t$ whenever $t = \frac{T}{2^n} i$ for some integer i, and linearly interpolated on intervals $[\frac{T}{2^n} i, \frac{T}{2^n}(i+1)]$.
Then, with probability one,

$$(B^{(n)}, \int_0^{\cdot} \otimes dB^{(n)}) \to (B, \mathbb{B}^{Strat}) \quad in \quad \mathscr{C}_g^{\alpha}.$$

(The integral uderstood as Riemann-Stieltjes integral.)

Brownian motion in a magnetic field

Brownian motion in a magnetic field

There exist smooth and uniform approximations to Brownian motion that do not converge to $(B, \mathbb{B}^{Strat})$, but to some different geometric rough path.

For example, consider

$$\bar{\mathbf{B}} = (B, \bar{\mathbb{B}}), \quad \textit{where} \quad \bar{\mathbb{B}}_{s,t} = \mathbb{B}^{Strat}_{s,t} + (t-s)A, \quad A \in \mathfrak{so}(d).$$

- The difference between $\bar{\mathbb{B}}$ and $\mathbb{B}^{Strat}$ is antisymmetric.
- Such approximations arise naturally.

Brownian motion in a magnetic field

A particle in $\mathbf{R}^3$ with mass m and position $x = x(t)$ with a nonzero electric charge is subject to

• frictions $\alpha_1, \alpha_2, \alpha_3 > 0$

• white noise in time, i.e. the distributional derivative of a Brownian motion B,

• magnetic field which is assumed to be constant.

$\Rightarrow$ By Newton's second law,

$$m\ddot{x} = -M\dot{x} + \dot{B}, \tag{6}$$

where $M \in \mathbf{R}^{d\times d}$ only has eigenvalues with positive real part.

Brownian motion in a magnetic field

$\Rightarrow$ Using $\quad p(t) = m\dot{x}(t)$,

$$\dot{p}(t) = -M\dot{x} + \dot{B} = -\frac{1}{m}M\dot{p} + \dot{B}. \tag{7}$$

Equivalently,

$$dX = \frac{1}{m}Pdt, \quad dP = -\frac{1}{m}MPdt + dB. \tag{8}$$

$\Rightarrow$ In this case, $X = X^m$ converges to $\bar{\mathbf{B}} = (B, \mathbb{B})$, where

$$\bar{\mathbb{B}}_{s,t} = \mathbb{B}^{Strat}_{s,t} + (t-s)A, \quad A \in \mathfrak{so}(d).$$

Brownian motion in a magnetic field

Theorem 3.8.

Let $M \in \mathbf{R}^{d\times d}$ be a square matrix in dimension d such that all its eigenvalues have strictly positive real part. Let B be a d-dimensional standard Brownian motion, $m > 0$, and consider the stochastic differential equations

$$dX = \frac{1}{m} P dt, \quad dP = -\frac{1}{m} MP dt + dB.$$

with zero initial position X and momentum P. Then, for any $q \geq 1$ and $\alpha \in (1/3, 1/2)$, as mass $m \to 0$,

$$\left(MX, \int MX \otimes d(MX) \right) \to \bar{\mathbf{B}} \quad \text{in } \mathscr{C}^\alpha \text{ and } L^q.$$

Scaling limits of random walks

Scaling limits of random walks

Consider continuous processes $\mathbf{X}^n = (X^n, \mathbb{X}^n)$, where

- values are in $V \oplus (V \otimes V)$, $dim V < \infty$,
- $\forall$ n, $\mathbf{X}^n_0 = (0, 0)$.

Theorem 3.10 (Kolmogorov tightness criterion for rough paths).

Let $q \geq 2$ and $\beta > 1/q$. For all s, t in $[0, T]$,

$$E_n(|X^n_{s,t}|^q) \leq C|t-s|^{\beta q}, \quad E_n(|\mathbb{X}^n_{s,t}|^{q/2}) \leq C|t-s|^{\beta q}, \tag{9}$$

for some constant $C < \infty$. Assume $\beta - 1/q > 1/3$.
Then, for all $\alpha \in (1/3, \beta - 1/q)$, the $\mathbf{X}^n$'s are tight in $\mathscr{C}^{0,\alpha}$.

Scaling limits of random walks

Proposition 3.11.

Consider a d-dimensional random walk $(X_j : j \in \mathbf{N})$, with i.i.d. increments of zero mean, finite moments of any order $q < \infty$, and unit covariance matrix. Extend the rescaled random walk

$$X^n_{\frac{j}{n}} := \frac{1}{\sqrt{n}} X_j,$$

defined on discrete times only, by piecewise linear interpolation to all times and construct to $\mathbf{X}^n = (X^n, \mathbb{X}^n)$ by iterated R-S integration. Then, the tightness estimates in Theorem 3.10 hold with $\beta = 1/2$ and all $q < \infty$.

Scaling limits of random walks

- $\mathbf{X}^n$ is a geometric rough path.
- $\mathbf{X}^n$ restricted to discrete times $\{\frac{j}{n} | j \in \mathbf{N}\}$ is a Lie group valued random walk.

$\Longrightarrow$ By central limit theorems on such Lie groups, $\mathbf{X}^n$ at unit time converges weakly to $\mathbf{B}^{Strat}$.

Scaling limits of random walks

Theorem 3.12.

In the rescaled random walk setting of Proposition 3.11, and under the additional assumption that $E(X \otimes X) = I$, we have the weak convergence

$$\mathbf{X}^n \Longrightarrow \mathbf{B}^{Strat}$$

in the rough path space $\mathscr{C}^\alpha([0,T]),\mathbf{R}^d)$, for any $\alpha < 1/2$.

Conclusion

Summary

1. For any $\alpha \in (1/3, 1/2)$ and $T > 0$, almost surely

$$\mathbf{B} = (B, \mathbb{B}) \in \mathscr{C}^{\alpha}([0, T], \mathbf{R}^d).$$
$$\mathbf{B} = (B, \mathbb{B}^{Strat}) \in \mathscr{C}_g^{\alpha}([0, T], \mathbf{R}^d).$$

2. Dyadic piecewise-linear approximations (B^n) to B satisfies a.s.,

$$(B^{(n)}, \int_0^{\cdot} \otimes dB^{(n)}) \to (B, \mathbb{B}^{Strat}) \quad in \quad \mathscr{C}_g^{\alpha}.$$

3. For any $q \geq 1$ and $\alpha \in (1/3, 1/2)$, as mass $m \to 0$,

$$\left(MX, \int MX \otimes d(MX)\right) \to \bar{\mathbf{B}} \quad \text{in } \mathscr{C}^{\alpha} \text{ and } L^q.$$

THANK YOU FOR LISTENING